UNIVERSITY OF WOLVERHAMPTON

Project Management 3.5

Business Measures

Terminology for Project Appraisal – measuring success

UNIVERSITY OF WOLVERHAMPTON

Learning Outcomes

- By the end of this week, you will be able to:
  - Estimate project costs and times
  - State why estimations are used
  - Discuss the differences between different estimation methods
  - Identify the need estimations
  - Give relevant estimation examples

Income

•For corporations, revenues minus cost of sales, operating expenses, and taxes, over a given period of time. Income is the reason corporations exist. Income is important to investors because they give an indication of the company's expected future dividends and its potential for growth and capital appreciation.
•Many young companies report negative income as they attempt to grow quickly enough to capture a new market, at which point they'll be even more profitable than they otherwise might have been. Also called earnings.

•For individuals, money earned through employment and investments.

Adapted from: http://www.investorwords.com/2400/income.html#ixzz1nPTCJgIK

Cash Flow

- To quote Charles Dickens' Mr Micawber addressing his wife in David Copperfield: **"Annual income twenty pounds, annual expenditure nineteen nineteen six, result happiness. Annual income twenty pounds, annual expenditure twenty pounds ought and six, result misery."**
- Cash flow: Incomings and Outgoings of cash, over the life of the project (e.g. by month, quarter or year)

- If you have more money going out than coming in, you have a problem

- Even if a company is profitable, if the cash flow is wrong, it can meet with financial disaster.

UNIVERSITY OF WOLVERHAMPTON

Capital Expenditure

Capital Expenditure (CapEx): Investments creating future benefits through the acquisition or upgrading of physical assets (e.g. buying a new machine)

UNIVERSITY OF WOLVERHAMPTON

Operational Expenditure

Operational Expenditure (OpEx): Expenses related to the production of its goods and services (e.g. salaries)

Return on Investment (ROI)

A performance measure used to evaluate the efficiency of an investment or to compare the efficiency of a number of different investments. To calculate ROI, the benefit (return) of an investment is divided by the cost of the investment; the result is expressed as a percentage or a ratio.The return on investment formula:

$$\text{ROI} = \frac{(\text{Gain from Investment} - \text{Cost of Investment})}{\text{Cost of Investment}}$$

Adapted from: http://www.investopedia.com/terms/r/returnoninvestment.asp#ixzz1nPWbUaS1

Return on Investment (ROI)

- Used or evaluating the financial consequences of investments.
- Compares the size and timing of investment gains directly with the size and timing of investment costs.

- The higher the ROI, the better!

$$\text{Simple ROI} = \frac{\text{Gains - Investment Costs}}{\text{Investment Costs}} = \frac{\$\ 700{,}000 - \$\ 500{,}000}{\$\ 500{,}000} = 40\%$$

Return on Investment (ROI)

and costs will appear as predicted.

- ROI as the **only** measure of an investment is flawed
  - It says nothing about the likelihood that expected returns and costs will appear as predicted. Good investment analysis will also measure the probabilities of different ROI outcomes so that an organisation can consider the ROI magnitude, the risks that go with it and the opportunities to improve ROI by changing or accelerating them



With ROI, decision makers compare the timing and magnitude of expected gains to investment costs.

UNIVERSITY OF WOLVERHAMPTON

Related Definitions

- **Sensitivities:** Analysis of the effect of changes to key assumptions in business economics

- **Key Performance Indicators:** The measures by which the success of the project will be measured (usually the addition to money); e.g. transaction time, headcount, customer satisfaction, CO2 emissions etc although these may have a cash value too.

UNIVERSITY OF WOLVERHAMPTON

Definitions

- Net Present Value (NPV)
  - The present value of an investment's future net cash flows minus the initial investment. If positive, the investment should be made (unless an even better investment exists), otherwise it should not
  - The difference between the present value of cash inflows and the present value of cash outflows. NPV is used in capital budgeting to analyse the profitability of an investment or project

UNIVERSITY OF WOLVERHAMPTON

How NPV is calculated

- n= positive integer
- N= total number of months/years
- C_n= Cash Flow
- r = Internal Rate of Return
- This MUST equal zero, or the investment should not be made (in theory)
- The higher the number, the better the value.

- A rate is usually in years, but should be in months if the project is of shorter duration – if longer, first months, and subsequently in years.

$$NPV = \sum_{n=0}^{N} \frac{C_n}{(1+r)^n} = 0$$

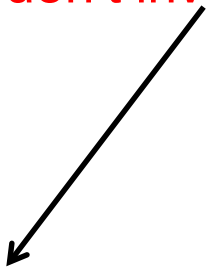
UNIVERSITY OF WOLVERHAMPTON

Net Present Value Example

Should you invest $60,000 in a project that will return $15,000 per year for five years? You have a minimum return of 8% and expect inflation to hold steady at 3% over the next five years.

Year	Net flow	Discount	NPV
0	-$60,000	1.0000	-$60,000.00
1	$15,000	0.9009	$13,513.51
2	$15,000	0.8116	$12,174.34
3	$15,000	0.7312	$10,967.87
4	$15,000	0.6587	$9,880.96
5	$15,000	0.5935	$8,901.77
			-$4,561.54

The NPV column total is negative, so don't invest!



Derived from Pinto, J. (2019) Project Management: Achieving Competitive Advantage, 5th Edition, Prentice Hall

UNIVERSITY OF WOLVERHAMPTON

Definitions

- Internal Rate of Return (IRR)
  - The discount rate often used in capital budgeting that makes the net present value of all cash flows from a particular project equal to zero.
  - Generally speaking, the higher a project's internal rate of return, the more desirable it is to undertake the project. IRR can be used to rank several prospective projects a firm is considering.
  - IRRs can also be compared against prevailing rates of return in the securities market. If a firm can't find any projects with IRRs greater than the returns that can be generated in the financial markets, it may simply choose to invest its retained earnings into the market

$$x_0 = 10\%$$

$$x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$$

UNIVERSITY OF WOLVERHAMPTON

IRR

- Internal refers to the fact that this does not take into consideration factors outside of the business itself – the economy, political environments, etc.
- This is not really a good measure by which to choose between two options, or projects of differing duration
- It is better to determine the decision based on a single project
- If one project has a higher investment cost than a second mutually exclusive project (that is, one or the other will be selected – not both), the first project may have a lower IRR (expected return), but a higher NPV (increase in shareholders' wealth) and should thus be accepted over the second project (assuming no capital constraints)

- Informed byhttp://en.wikipedia.org/wiki/Internal_rate_of_return

UNIVERSITY OF WOLVERHAMPTON

IRR

- This makes the Net Present Value of a project equal to zero when calculated, so a measure of improvement can be made
- A discount rate where Net Present Value of Costs = Net Present Value of Benefits
- This helps assess how long the investment will take to break even
- it is an indicator of the efficiency, quality, or yield of an investment.

UNIVERSITY OF WOLVERHAMPTON

How IRR is calculated

- The internal rate of return follows from the net present value as a function of the rate of return. A rate of return for which this function is zero is an internal rate of return.

year 0 + year1 + year2 + year3

In this case, the answer is 5.96% (in the calculation, that is, r = .0596).

Year (n)	Cash Flow (C_n)
0	-123400
1	36200
2	54800
3	48100

$$NPV = -123400 + \frac{36200}{(1+r)^1} + \frac{54800}{(1+r)^2} + \frac{48100}{(1+r)^3} = 0$$

UNIVERSITY OF WOLVERHAMPTON

Internal Rate of Return Example

A project that costs $40,000 will generate cash flows of $14,000 for the next four years. You have a rate of return requirement of 17%; does this project meet the threshold?

Year	Net flow	Discount	NPV
0	-$40,000	1.0000	-$40,000.00
1	$14,000	0.9009	$12,173.91
2	$14,000	0.8116	$10,586.01
3	$14,000	0.7312	$9,205.23
4	$14,000	0.6587	$8,004.55
			-$30.30

This table has been calculated using a discount rate of 15%

The project doesn't meet our 17% requirement and *should not be considered further*.

Derived from Pinto, J. (2019) Project Management: Achieving Competitive Advantage, 5th Edition, Prentice Hall

UNIVERSITY OF WOLVERHAMPTON

Definitions

- Payback period
  - How long does it take for an investment to pay for itself?
  - How long does it take for incoming returns to cover costs?
  - How long does it take for the investment to break even?

Payback Period = Cost of Project / Annual Cash Inflows

UNIVERSITY OF WOLVERHAMPTON

Payback Period Example

A project requires an initial investment of $200,000 and will generate cash savings of $75,000 each year for the next 3 years. What is the payback period?

Year	Cash Flow	Cumulative
0	($200,000)	($200,000)
1	$75,000	($125,000)
2	$75,000	($50,000)
3	$75,000	$25,000

Divide the cumulative amount by the cash flow amount in the third year and subtract from 3 to find out the moment the project breaks even.

$$3 - \frac{25,000}{75,000} = 2.67 \ years$$

Derived from Pinto, J. (2019) Project Management: Achieving Competitive Advantage, 5th Edition, Prentice Hall

UNIVERSITY OF WOLVERHAMPTON

Financial and Economic

- Financial considerations:
  - Financial viability of the project
  - Adequacy of funds
  - Ability to service organisational debt for the project
  - How to recover investment
  - Operating costs

  Bottom line:
  - Does the project generate a profit for the organisation?

UNIVERSITY OF WOLVERHAMPTON

Financial and Economic

- Economic considerations:
  - Is the project worthwhile by other measures?
    - Social Welfare
    - Aggregate organisational good
    - National good
    - Cultural good
    - Customer experience

  Bottom line:
  - Does the project generate benefits for the organisation?

UNIVERSITY OF WOLVERHAMPTON

Project Appraisal

- There are many ways to determine value
  - What is right is the one that will give the most accurate view of the organisational circumstances
  - Many methodologies address this directly, and supply templates and other measures
  - All of these are tools to help decision making – nothing can replace common sense in these environments.